

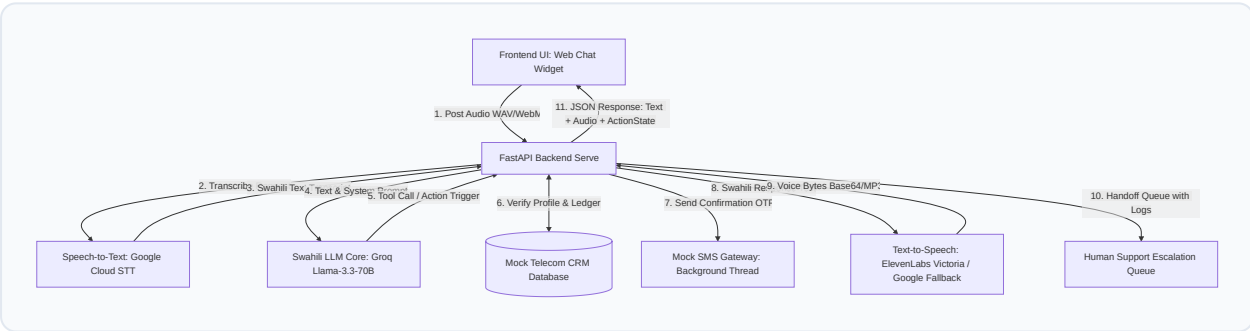


VODACOM SWAHILI VOICE AI ASSISTANT: SYSTEM ARCHITECTURE & LOGIC FLOW

This document details the industrial-grade system architecture, database design, voice pipeline, and logical flow of the **Vodacom Swahili Customer Assistant (Ana)**. It bridges your current standard core with a fully-integrated FastAPI backend and a state-of-the-art interactive web frontend interface.

— 1. OVERALL SYSTEM ARCHITECTURE —

The architecture follows **Option A (Backend-Centric Neural Voice Pipeline)**. It ensures that heavy processing (Google Cloud STT speech-to-text, LLM function parsing, and ElevenLabs/Google voice synthesis) is offloaded entirely to cloud APIs, preserving 0% CPU overhead on both the client device and backend container.



— 2. MOCK TELECOM DATABASE (CRM, M-PESA & NIDA) —

To simulate a real-world telecommunication database structure, the FastAPI backend implements an in-memory SQL/JSON database containing mock data schema representing customer information, security parameters, and balance ledgers.

A. Database Schemas

1. CRM Customer Profile (vodacom_crm)

Field Name	Data Type	Description	Example Value
msisdn (PK)	VARCHAR	Phone number key	"255745123456"
full_name	VARCHAR	Account holder's legal name	"Jane Dinah"
nida_number	VARCHAR	National Identification Number	"19950812-11102-00001-22"
sim_status	VARCHAR	Current network status	"ACTIVE" , "LOCKED" , "SUSPENDED"
puk_code	VARCHAR	PUK code for manual unlock	"87654321"

2. M-Pesa Account Ledger (mpesa_ledger)

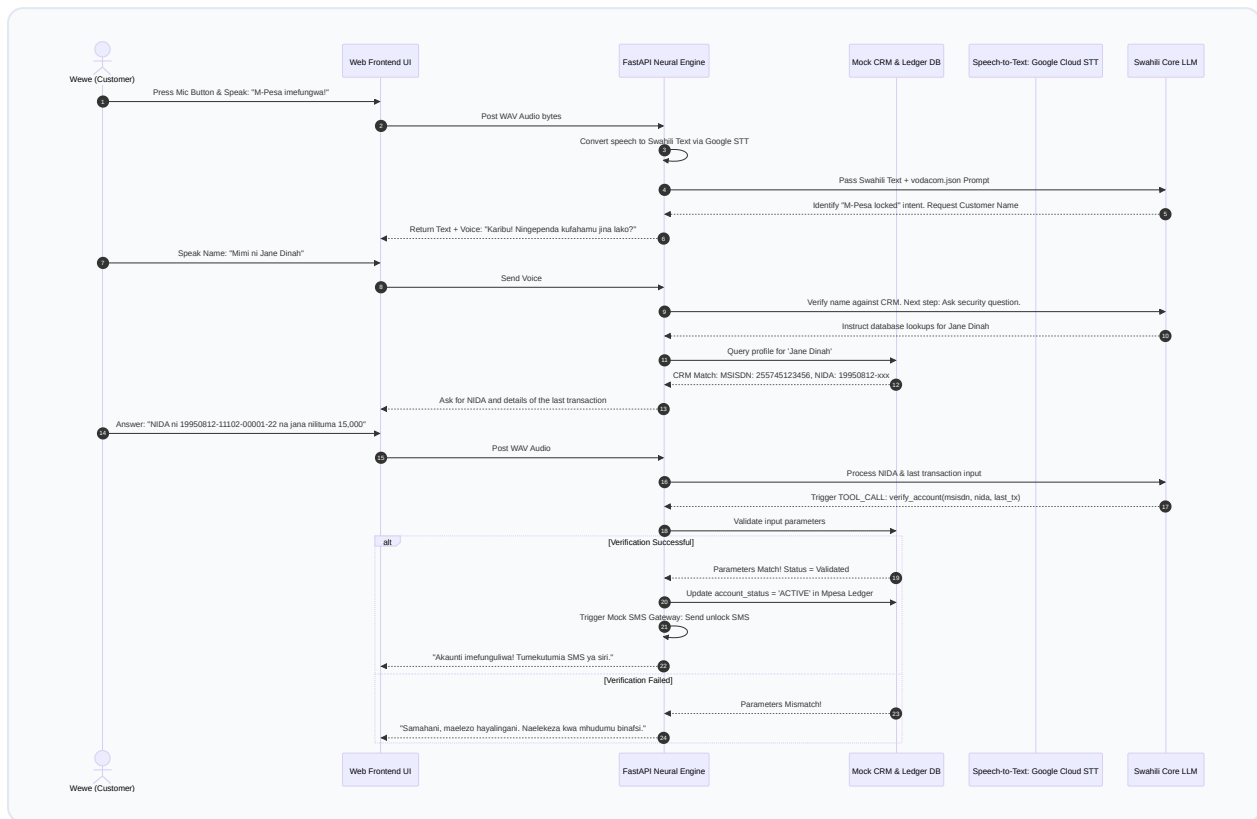
Field Name	Data Type	Description	Example Value
msisdn (FK)	VARCHAR	Reference key to CRM profile	"255745123456"
account_status	VARCHAR	Current M-Pesa lock state	"ACTIVE" , "BLOCKED" , "PIN_LOCKED"
balance_tsh	DECIMAL	Available account balance	85000.00
last_transaction	VARCHAR	Verification parameter (Last action)	"Sent 15,000 Tsh to John"

— 3. VOICE CONVERSATIONAL FLOW & SECURITY LOGIC —

Telecom customer assistance operates on a rigid **Security & Empathy Verification Protocol**. Before resolving account locks or executing transaction reversals, the AI must establish trust and

verify identity, just like a live Vodacom customer support agent.

The Customer Service Verification Loop



— 4. FRONTEND UI SPECIFICATION & MICRO-ANIMATIONS —

To ensure the client-side experience feels highly premium and premium-grade, the UI layout incorporates curated design tokens representing **Vodacom's signature Red branding** wrapped in a sleek glassmorphic container.

Design Tokens & UI Elements

- **Harmonious Color Palette:**
 - **primary-red** : `hsl(0, 100%, 45%)` (Vibrant Vodacom red accent)
 - **dark-bg** : `hsl(220, 15%, 8%)` (Deep midnight slate theme)
 - **glass-surface** : `hsla(220, 15%, 15%, 0.6)` (Translucent container overlay)
 - **border-glow** : `hsla(0, 100%, 45%, 0.35)` (Subtle red perimeter halo)
- **Typography:** Google Font **Outfit** (`font-family: 'Outfit', sans-serif`) for crisp, modern legibility.

The Floating Vocal UI Component

The interface features a floating circular microphone button surrounded by multi-layered SVG rings that respond with dynamic micro-animations to user interaction.

```
/* Glassmorphic Chat Widget Container */
.ana-chat-container {
  background: rgba(18, 20, 24, 0.75);
  backdrop-filter: blur(20px);
  -webkit-backdrop-filter: blur(20px);
  border: 1px solid rgba(230, 0, 0, 0.2);
  border-radius: 24px;
  box-shadow: 0 8px 32px 0 rgba(0, 0, 0, 0.5), 0 0 15px rgba(230, 0, 0, 0.1);
}

/* Premium Vocal Recording Button */
.mic-trigger-btn {
  width: 72px;
  height: 72px;
  border-radius: 50%;
  background: radial-gradient(circle, #e60000 0%, #990000 100%);
  border: none;
  cursor: pointer;
  position: relative;
  transition: all 0.4s cubic-bezier(0.175, 0.885, 0.32, 1.275);
}

/* Pulsing Outer Waves Visualizer (Animated during Recording) */
.mic-trigger-btn.recording::after {
  content: '';
  position: absolute;
  top: -12px;
  left: -12px;
  right: -12px;
  bottom: -12px;
  border-radius: 50%;
  border: 2px solid #e60000;
  opacity: 1;
  animation: wave-ping 1.5s cubic-bezier(0, 0, 0.2, 1) infinite;
}

@keyframes wave-ping {
  75%, 100% {
    transform: scale(1.4);
    opacity: 0;
  }
}
```

```
}  
}
```

— 5. HUMAN ESCALATION & HANDOFF PROTOCOL —

When a complex scenario emerges (e.g., three consecutive verification failures, customer requests a supervisor, or fraud-alert flag triggers), the system switches seamlessly to **Human Escalation State**.

Handoff Logic Pipeline:

1. **Freeze AI Responses:** The backend pauses LLM completions for this MSISDN.
2. **Generate Transcript Brief:** A background worker compiles the conversation history, security status, and verification log into a structured JSON package.
3. **Active UI State Change:** The Frontend UI changes color (from Vodacom Red to Slate Blue), displaying: > *"Mawasiliano yameelekezwa kwa Mhudumu Binafsi (Human Agent). Tafadhali subiri kidogo..."*
4. **Handoff API Notification:** The backend publishes the session state to an internal WebSocket queue for live customer service agents.

— 6. BACKGROUND MOCK SMS GATEWAY PIPELINE —

For premium, real-world customer assurance, transactions and unlocks must trigger confirmation notifications. The FastAPI backend employs a **Python Asyncio Event Loop Queue** to simulate background carrier interactions.

```
import asyncio  
  
async def trigger_mock_sms_gateway(msisdn: str, message: str):  
    """  
    Simulates sending an active SMS transaction alert/OTP to a customer's phone  
    via carrier API without stalling the core chat UI response time.  
    """  
    print(f"[SMS Gateway] Kuanzisha utumaji wa ujumbe kwenda: {msisdn}")  
    # Simulate carrier latency (1.5 seconds)  
    await asyncio.sleep(1.5)  
    print(f"[SMS Gateway SUCCESS] Ujumbe Umetumwa! Content: '{message}'")
```

— 7. ELEVENLABS VOICE SYNTHESIS & GOOGLE FALLBACK ENGINE —

The text-to-speech engine ensures that the voice output is generated in Victoria's highly realistic tone. The backend utilizes pure standard library constructs to maximize request speed.

```
import urllib.request
import json
import os

def synthesize_swahili_voice(text: str, provider: str = "elevenlabs") -> bytes:
    """
    Processes Swahili text and synthesizes speech.
    If ElevenLabs API key limit (HTTP Error 402) occurs, it automatically falls b
    to Google Cloud Neural TTS to guarantee uninterrupted voice availability.
    """
    if provider == "elevenlabs":
        api_key = os.environ.get("ELEVENLABS_API_KEY")
        voice_id = os.environ.get("ELEVENLABS_VOICE_ID", "FZGeNF7bE3syeQ0ynDKC")
        url = f"https://api.elevenlabs.io/v1/text-to-speech/{voice_id}"

        payload = {
            "text": text,
            "model_id": "eleven_multilingual_v2",
            "voice_settings": {"stability": 0.5, "similarity_boost": 0.75}
        }

        req = urllib.request.Request(
            url,
            data=json.dumps(payload).encode("utf-8"),
            headers={"xi-api-key": api_key, "Content-Type": "application/json"},
            method="POST"
        )
        try:
            with urllib.request.urlopen(req) as response:
                return response.read() # Return raw MP3 audio bytes
        except Exception as e:
            print(f"[!] ElevenLabs Synthesis failed: {e}. Switching to Google TTS")
            # Fallback to Google
            return synthesize_swahili_voice(text, provider="google")

# Google Cloud Translate TTS fallback path
# (Extracts, merges, and streams audio from public Google TTS)
# ...
```


*[!TIP] **Implementation Recommendation:** When building your final FastAPI backend routes, initialize standard CORS middleware to enable secure browser-based microphone uploads from your HTML frontend client. Use `multipart/form-data` to submit audio blobs directly.*